August 5, 2026

Editor-in-Chief
Journal of Systems and Software

Dear Editor,

Subject: Submission of manuscript "Recovery-Path Trust Boundaries in Self-Healing LLM Agent Systems: Architectural Analysis, Empirical Evidence, and Design Patterns"

We are pleased to submit the above-referenced manuscript for consideration by the *Journal of Systems and Software*.

Summary. This paper studies the architectural trust assumptions embedded in self-healing LLM agent systems—automated recovery pipelines that capture execution failures and re-inject diagnostic context into subsequent agent iterations. We identify a systematic architectural anti-pattern (the *trust-escalation trampoline*) in which the recovery path, designed to improve robustness, bypasses the provenance and privilege invariants enforced on the forward execution path. We propose two OS-principled design patterns—Provenance Tagging at capture sites and a Privilege Downgrade Invariant at a centralized reauthorization gate—and validate their implementability and efficacy on a white-box testbed.

Contribution type. This is a software architecture and design contribution. The paper focuses on: (i) formalizing the recovery-path trust boundary as an architectural concept, (ii) proposing two reusable design patterns with OS analogies (provenance tagging, privilege containment), and (iii) providing a concise validation that the patterns are implementable with zero false alarms and sub-millisecond overhead. The paper deliberately does *not* include the full statistical evaluation (attack success rates with confidence intervals, cross-model generalization, defense-aware attackers, production-framework exploits)—that material is reported in a companion empirical study.

Disclosure of companion manuscript. We wish to transparently disclose that the authors have prepared a companion manuscript submitted to *Empirical Software Engineering* (EMSE), titled "Trust Assumptions in Self-Healing LLM Agent Frameworks: An Empirical Code Analysis of Recovery-Path Provenance Gaps." The two manuscripts are complementary and non-overlapping in their primary contributions:

- **This manuscript (JSS)** contributes the *architectural design*: the formalization of the trust-escalation trampoline anti-pattern, the two design patterns (Provenance Tagging + Privilege Downgrade Invariant), and a concise validation of implementability.
- **The companion manuscript (EMSE)** contributes the *empirical evaluation*: a multiple-case study of six production agent frameworks, controlled experiments with statistical tests (Wilson CIs, Fisher exact tests, inter-annotator agreement), a confirmed end-to-end exploit on the Reflexion framework, cross-model generalization, and defense-aware attacker analysis.

The two manuscripts cite each other and are each self-contained: a reader of either paper can understand the full contribution without accessing the other. We have taken care to ensure that textual overlap between the two manuscripts is minimal (estimated <15%); each manuscript uses its own language to describe the shared system and attack surface.

We are happy to provide the companion manuscript as a confidential attachment for the editor's review, should that be helpful for verifying non-overlap. We respectfully request that the companion manuscript not be shared with reviewers, to preserve the double-blind integrity of both submissions.

Fit with JSS. The manuscript fits JSS's scope in software architecture, design patterns, and system design: it identifies an architectural anti-pattern in an emerging system class (self-healing LLM agents), proposes reusable design patterns grounded in classical OS principles, and validates their implementability. The full empirical evaluation, which uses security-research methodologies (ASR statistics, red-team experiments), is more appropriate for EMSE's empirical-software-engineering audience.

We confirm that this manuscript has not been published and is not under consideration elsewhere. We look forward to the editors' and reviewers' feedback.

Sincerely,

Anonymous Author(s)
(for double-blind review)